

PROGRESS REPORT

Hi-CAS: Helmet-Integrated Conformal Antenna System

Field	Details
Problem Statement ID	26185
Problem Statement Title	Helmet mounted conformal antenna for tactical communications in urban CQB environments
Theme	Robotics and Drones
PS Category	Hardware
Team ID	007
Team Name	BLACKCATS
Team Motto	One Team. One Link. Zero Communication Gaps.
Report Date	6 September 2026

Status: Concept design, RF stack-up, mechanical CAD and preliminary structural simulation complete.

1. Executive Summary

Team BLACKCATS is developing Hi-CAS (Helmet-Integrated Conformal Antenna System), a low-profile, helmet-conformal dual-band antenna solution that replaces the traditional external whip antenna used in tactical radios. This report summarizes the work completed to date against Problem Statement 26185: concept definition, RF architecture and layer stack-up design, patch antenna sizing and substrate selection, helmet CAD modelling in Fusion 360, and a first-pass structural (static stress) simulation of the antenna-integrated helmet shell under a 600 N load.

At this stage, the design and its RF/mechanical rationale are complete and documented; the structural simulation shows a comfortable safety margin (minimum safety factor 4.72) and negligible deformation, indicating that the added antenna layers do not compromise the mechanical integrity of the underlying helmet shell in this simplified static model. The full layered helmet CAD model (all stack-up layers at their specified dimensions) has also been successfully constructed and simulated in Fusion 360. On the RF side, construction of the L-band antenna has begun: the team started with a planar (flat) patch element before attempting a curved/bent version, and CST simulation — after initially returning a flat, resonance-free S-parameter line — now shows clear resonant dips, confirming the element is functioning and ready for dimensional tuning. Detailed EM simulation of the conformal/curved arrays, physical prototyping, and ballistic-grade validation remain as next steps.

2. Problem Statement

Traditional external whip antennas mounted on tactical helmets create a physical and RF trade-off in confined, close-quarter-battle (CQB) urban environments:

- Exposed hardware: external whip antennas protrude and snag on surroundings, interfering with operator movement.
- Non-conformal form: rigid antennas do not follow helmet geometry, causing fit and integration issues.
- RF performance variation: helmet curvature, proximity to human tissue, and the surrounding environment affect antenna impedance and radiation behaviour.
- Multi-band challenge: a single compact platform must reliably support and operate across both UHF and L-band tactical frequencies.

3. Proposed Solution — Hi-CAS

Hi-CAS integrates two independent, low-profile conformal antenna arrays onto the helmet surface, connected through a diplexer so that both bands share a single RF interface to the radio:

- UHF array (200–450 MHz): a conformal array optimized for the UHF tactical band, positioned on the rear/crown region of the helmet.
- L-band array (1.2–1.5 GHz): a second conformal array optimized for the L-band, positioned separately from the UHF array to minimize mutual coupling.
- 3-port diplexer: combines/separates the UHF and L-band signals into a single RF interface to the transceiver.
- Phase-control network: a phase shifter optimizes signal coverage for reliable, $\sim 360^\circ$ azimuth tactical communication.

THE PROBLEM → OUR SOLUTION

Traditional external antennas create a physical and RF trade-off in confined urban environments.

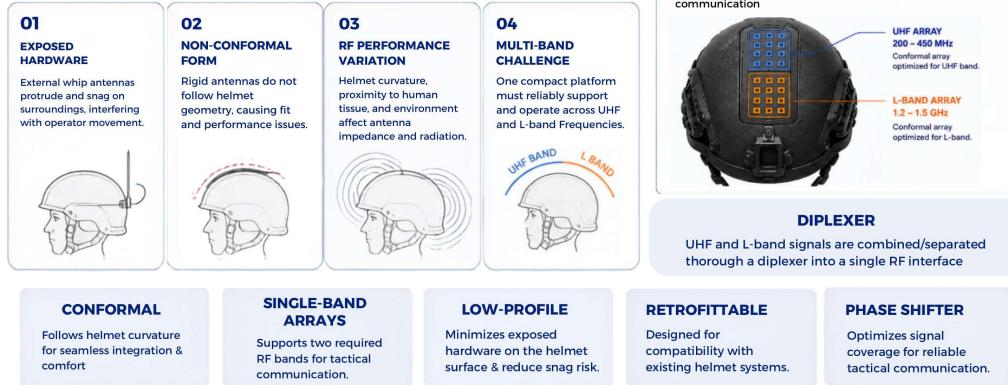


Figure 1: Hi-CAS solution overview — from exposed external antenna to an integrated, helmet-conformal dual-band system (Team BLACKCATS, SIH 2026).

Design pillars carried through the concept:

- Conformal — follows helmet curvature for seamless integration and comfort.
- Single-band arrays — two purpose-built arrays instead of one compromised multi-band element.
- Low-profile — minimizes exposed hardware on the helmet surface and reduces snag risk.
- Retrofittable — designed for compatibility with existing helmet systems.
- Phase-shifted feed — optimizes coverage for reliable tactical communication.

4. RF and Communications Architecture

4.1 End-to-end communication stack

The antenna is the physical layer of a complete tactical communications stack. The system architecture defined so far is:

- Helmet Antenna (Hi-CAS conformal arrays)
- RF Transceiver / Radio
- PHY Layer
- Data Link Layer
- Networking Layer
- Transport Layer
- Application Layer — tactical COMSEC / TRANSEC implementation

4.2 Antenna design and validation workflow

The antenna element itself is being developed and validated in stages, moving from a simple flat patch to a full conformal, multi-element array:

- Design a single microstrip patch element.
- Check S11 / VSWR.
- Check the 3D radiation pattern.
- Rotate the antenna and evaluate different orientations.
- Understand how the radiation pattern changes with orientation.
- Add helmet curvature to the model (transition from flat to conformal).

- Develop multiple conformal elements.
- Combine elements into arrays.
- Optimize the combined arrays for close to 360° azimuth coverage.

4.3 Patch antenna element and substrate

The radiating element follows the classical microstrip patch configuration — a copper patch on a grounded dielectric substrate, fed by a microstrip feed line.

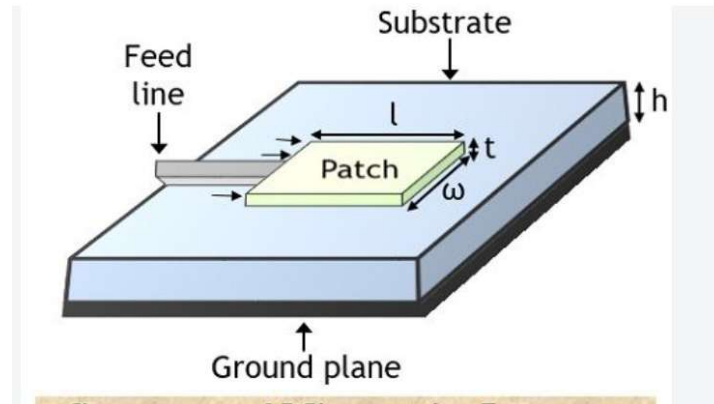


Figure 2: Generic microstrip patch antenna geometry — patch, feed line, dielectric substrate and ground plane.

RT/duroid 5880 was selected as the working substrate for initial sizing and simulation because of its low loss tangent and stable dielectric constant, which suit conformal antenna applications:

Parameter	RT/duroid 5880
Substrate	RT/duroid 5880
Substrate thickness	0.127 mm
Copper thickness	35 μm ($\sim 1 \text{ oz/ft}^2$)
Antenna conductor	Copper
Ground	Copper

4.4 Helmet sizing constraints on antenna footprint

Antenna footprint was cross-checked against the Ministry of Home Affairs Bullet Resistant Helmet specification, which defines head circumference and sagittal-arc ranges for three helmet sizes. The sagittal arc — the curved distance over the top of the head from the glabella to the base of the occiput — is the most relevant dimension for conformal antenna placement.

Size	Head circumference	Sagittal arc
Small	48–52 cm	30–35 cm
Medium	>52–57 cm	>35–39 cm
Large	>57–64 cm	>39–44 cm

Note: the sagittal arc describes the helmet/head curvature, not the available flat antenna footprint. For a medium helmet, the working assumption for the central curved patch region is approximately 120–160 mm long $\times$ 70–100 mm wide, with exact dimensions to be finalized through HFSS/CST optimization for the chosen UHF and L-band frequencies. As a cross-reference, commercial Indian tactical helmets (e.g. MKU KAVRO DOMA 360) specify head circumferences of 500–530 mm, 530–570 mm and 570–600 mm for small, medium and large respectively.

4.5 Conformal layer stack-up

Two candidate stack-ups have been worked through for the flexible antenna assembly bonded to the helmet shell. The first is an early dimensional draft (each layer cut progressively larger than the one below it, from radiator outward, to allow for lamination tolerance):

Layer	Starting length	Starting width	Thickness
Radiating copper	150 mm	90 mm	35 μm
Flexible substrate	160 mm	100 mm	0.127 mm / 0.05 mm
Ground copper	155 mm	95 mm	35 μm
Kapton isolation	160 mm	100 mm	50 μm
Ferrite absorber	160 mm	100 mm	100–200 μm
Protective outer film	165 mm	105 mm	application-dependent

This was consolidated into the current working stack-up (outside to inside), which is the version carried into the helmet integration model:

Order (outside $\rightarrow$ inside)	Layer	Thickness
1	Silicone rubber protective layer	—
2	Copper radiator	0.035 mm
3	RT/duroid 5880 (dielectric)	0.127 mm
4	Copper ground	0.035 mm
5	Kapton isolation	0.05 mm
6	Ferrite absorber (if retained)	0.20 mm
7	Helmet shell	15 mm

A key design rule carried through both iterations: the ballistic helmet shell remains structurally uninterrupted. The full functional stack, from outside to head, is: thin protective coating $\rightarrow$ flexible antenna (RF radiating layer) $\rightarrow$ flexible dielectric $\rightarrow$ RF ground/shield $\rightarrow$ electrical isolation $\rightarrow$ ballistic helmet shell (bullet protection, uninterrupted) $\rightarrow$ impact-absorbing liner $\rightarrow$ head. No RF layer penetrates or replaces load-bearing ballistic material.

4.6 L-band Antenna — First EM Simulation Results

Construction of the L-band antenna has begun. Following the staged workflow in Section 4.2, the team deliberately started with a flat (planar) patch antenna before attempting the curved/bent conformal version, so that the basic resonance behaviour of the element could be verified on simple, well-understood geometry first.

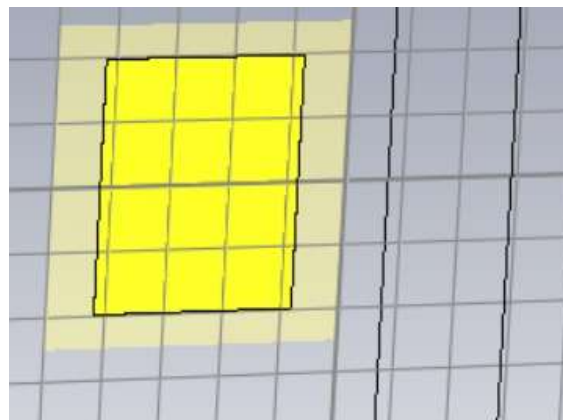


Figure 5: Planar L-band patch element modelled on the CST mesh grid — the starting geometry before curvature/bending is introduced.

The first simulation attempt on this planar patch returned a flat S1,1 line with no visible resonance, indicating an issue with the initial port/feed or dimension setup. After revisiting the model, a re-run produced a proper S-parameter response with multiple resonance dips across the swept band:

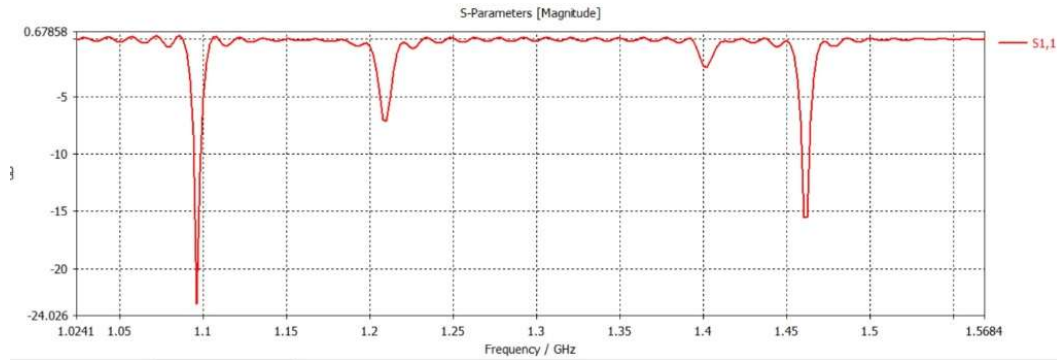


Figure 6: $S_{1,1}$ magnitude vs. frequency for the planar L-band patch after correction — resonances near 1.10 GHz (-24.0 dB), 1.21 GHz (-7.1 dB), 1.40 GHz (-3.0 dB) and 1.46 GHz (-15.3 dB).

This confirms the patch is resonating and that the simulation setup (port, mesh, boundary conditions) is now functioning correctly. The strongest, cleanest dip at present falls just below the intended L-band (near 1.10 GHz), with a secondary, shallower dip at 1.46 GHz inside the 1.2–1.5 GHz target band. The next iteration will retune the patch length/width and feed position to consolidate this into a single, deep resonance centred inside the target L-band, before repeating the process on the curved/conformal version of the element.

5. Mechanical Design (CAD)

The helmet shell and the conformal antenna assembly were modelled together in Autodesk Fusion 360 as a multi-body assembly and exported as a STEP AP214 file (Basic_helmet_element_design.step) for cross-platform CAD exchange and downstream simulation.

- The model is a dome-shaped ballistic helmet shell with a curved patch region on the crown representing the conformal antenna patch/array footprint.
- The assembly is decomposed into 14 solid bodies, matching the distinct material layers of the stack-up (helmet shell, liner, dielectric substrate, copper radiator, copper ground, protective/isolation layers).
- This body-per-layer approach allows each layer to be assigned its own material properties for structural simulation, and keeps the RF layer stack traceable back to the CAD model.
- The basic helmet design was successfully built and simulated in Fusion 360 with all stack-up layers present at their specified dimensions (Section 4.5), confirming the layered geometry is structurally sound and simulation-ready before EM co-simulation is attempted.

6. Structural Simulation — Static Stress Study (600 N)

A first-pass static structural simulation was run on the assembled CAD model in Fusion 360 to check whether bonding the thin antenna layers onto the helmet shell introduces any structural weakness, and to establish a baseline load-response before ballistic-specific testing.

6.1 Study setup

Setting	Value
Study type	Static Stress
Analyzed file	helmet (Fusion 360, v2704.1.53)
Mesh type	Solids, parabolic elements, curved mesh enabled
Average element size	10% of model size
Nodes / Elements	946,653 nodes / 469,752 elements
Constraint	Fixed support at the helmet rim (all translations fixed)
Applied load	Two 600 N forces on the crown/antenna patch region (representing an impact-type load case), oriented predominantly downward
Results basis	Baseline accuracy, von Mises stress criterion

6.2 Materials assigned per layer

Body / layer	Material	Yield strength	Young's modulus
Body1, 8–14 (shell/structure)	Nylon 6/6 (variant 4)	2920.00 MPa*	70,300 MPa
Body2 (shell layer)	Nylon 6/6 (variant 3)	45.00 MPa	120,000 MPa
Body3 (liner)	Nylon 6/6 (variant 2)	69.00 MPa	2,500 MPa
Body4, 6 (radiator / ground)	Copper	33.30 MPa	117,500 MPa
Body5 (dielectric)	Nylon 6/6 (variant 1)	138.00 MPa	3,712 MPa
Body7 (protective layer)	TPU	19.00 MPa	1,500 MPa

*As exported from the material library for this body; to be re-checked against manufacturer datasheets before the next simulation pass, since it is high relative to typical Nylon 6/6 yield values.

6.3 Load and boundary condition

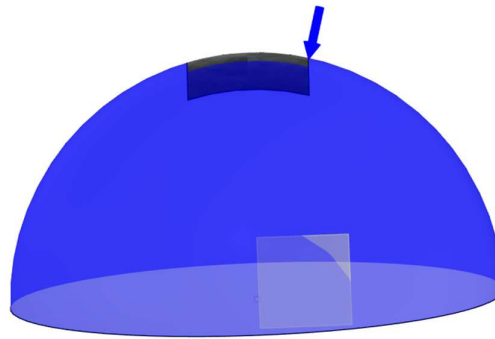


Figure 3: Load case — 600 N applied on the crown/antenna-patch region, with the helmet rim fixed.

6.4 Key results

Result	Minimum	Maximum
Safety factor	4.72	> 500,000 (capped)
Von Mises stress	~0 MPa	7.06 MPa
1st principal stress	-5.20 MPa	13.81 MPa
3rd principal stress	-9.76 MPa	6.69 MPa
Total displacement	0 mm	0.001 mm
Total reaction force	0 N	4.68 N
Equivalent strain	1.03×10^{-7}	1.57×10^{-4}
Contact pressure	0 MPa	13.77 MPa

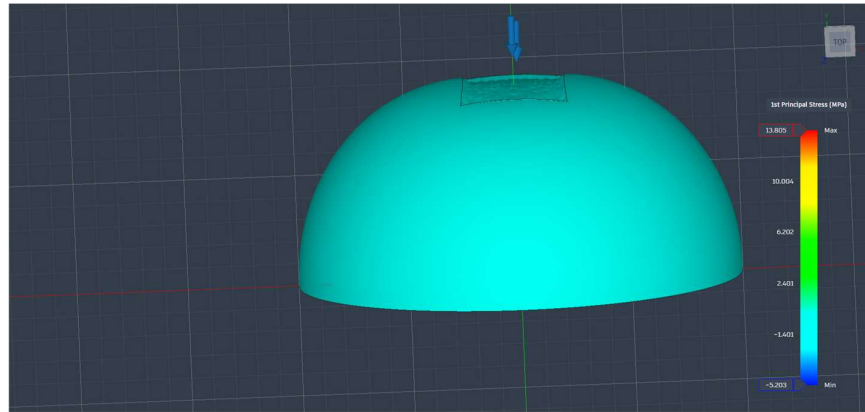


Figure 4: 1st principal stress distribution across the helmet shell under the 600 N load case (max 13.81 MPa, localized at the antenna patch region).

Interpretation: the minimum safety factor of 4.72 is well above 1.0, indicating that, in this simplified static model, none of the layers — including the copper radiator/ground and dielectric — is predicted to yield under the applied 600 N load. Displacement is negligible (≤ 0.001 mm), and peak stresses remain concentrated at the antenna patch region and well below the yield strengths of the assigned materials. This is a preliminary structural sanity check, not a certified ballistic test: it confirms that bonding the thin (sub-millimetre) antenna stack onto the shell does not introduce an obvious structural weak point, but it does not replace dedicated ballistic impact testing.

7. Feasibility and Engineering Challenges

Core antenna and helmet technologies already exist individually; the innovation and remaining risk lie in integrating them and validating performance under realistic conditions. Four challenge areas have been identified, each with a defined measurement approach:

Challenge	Description	How it will be measured
Curvature	Bending the antenna to follow the helmet surface alters its electrical length and EM coupling.	Resonant frequency, S11, VSWR, bandwidth (flat vs. conformal).
Temperature	Environmental temperature variation affects material and antenna behaviour.	RF performance before and after temperature cycling.
Head proximity	The operator's head changes the EM environment around the antenna.	Impedance, efficiency, and radiation-pattern characteristics with/without a head phantom.
Structural integration	The antenna must not compromise helmet functionality or ballistic protection.	Integration compatibility and structural/ballistic checks (no certification claimed at this stage).

7.1 Mitigation and validation sequence

The team is following a seven-step sequence to move from concept to a validated prototype:

- 1. Design
- 2. Simulate
- 3. Prototype
- 4. Bend test
- 5. Temperature test
- 6. RF validation
- 7. Structural validation

8. Progress Summary — What Has Been Done So Far

Area	Status	Output produced
Problem framing & solution concept	Complete	Hi-CAS dual-band conformal architecture defined (UHF + L-band arrays, diplexer, phase control).
Communications architecture	Complete (concept)	Full stack from antenna to application layer defined; per-band frequency ranges fixed (UHF 200–450 MHz, L-band 1.2–1.5 GHz).
Antenna element design workflow	In progress	9-step validation workflow defined; L-band planar patch built and simulated in CST with resonance confirmed (Section 4.6).
Substrate & material selection	Complete (initial)	RT/duroid 5880 dielectric, copper radiator/ground, Kapton isolation, ferrite absorber, silicone protective layer.
Layer stack-up dimensioning	Complete (initial), consolidated	Two iterations of the stack-up produced and reconciled into one working stack.
Helmet sizing cross-check	Complete	Antenna footprint checked against MHA helmet-size spec and a commercial reference helmet.
Mechanical CAD (helmet + antenna layers)	Complete (v1)	14-body Fusion 360 assembly with all stack-up layers built to specified dimensions and successfully simulated; exported as STEP (Basic_helmet_element_design.step).
Structural simulation	Complete (v1, static, 600 N)	Static stress study: min. safety factor 4.72, max. displacement 0.001 mm, max. von Mises stress 7.06 MPa.
Electromagnetic simulation (S11/VSWR, pattern, gain)	Started (L-band, planar)	Planar L-band patch simulated in CST Studio Suite; resonance behaviour confirmed, dimensions being tuned to centre on target band.
Physical prototyping & bend/temperature testing	Not started	Planned after EM simulation is complete.

9. Impact and Potential Applications

Impact area	Benefit
Operational	Reduced physical obstruction and improved operator mobility.
RF performance	Independent, band-optimized arrays combined via a diplexer.
User safety	Fewer exposed antenna parts, reducing snag and injury risk.
Scalability	Modular UHF/L-band architecture for broad helmet and radio compatibility.

Potential end users: NSG / ATS teams and other specialized tactical units, and other helmet-based communication systems, subject to further validation and procurement processes.

10. Next Steps

- Run full electromagnetic simulation of the single patch and the combined conformal arrays in CST Studio Suite (S11/VSWR, 3D radiation pattern, gain, RF isolation between UHF and L-band arrays).
- Finalize the exact patch and array dimensions for the chosen UHF and L-band frequencies, replacing the current ~120–160 mm × 70–100 mm working estimate.
- Re-verify assigned material properties (especially the high yield-strength value flagged in Section 6.2) against manufacturer datasheets.
- Fabricate a flexible-PCB prototype of the antenna stack-up and bond it to a test helmet shell section.

- Execute the bend test, temperature-cycling test, RF validation, and structural validation stages of the mitigation sequence.
- Re-run the structural simulation with prototype-derived material data and, where possible, a more representative (dynamic/impact) load case.

— *End of report* —